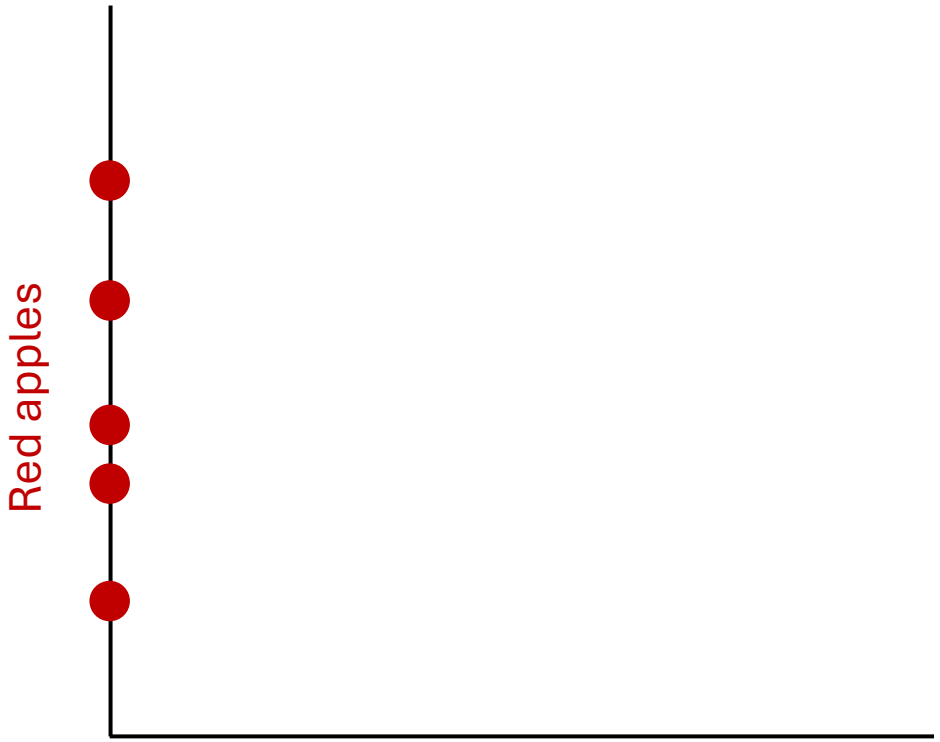
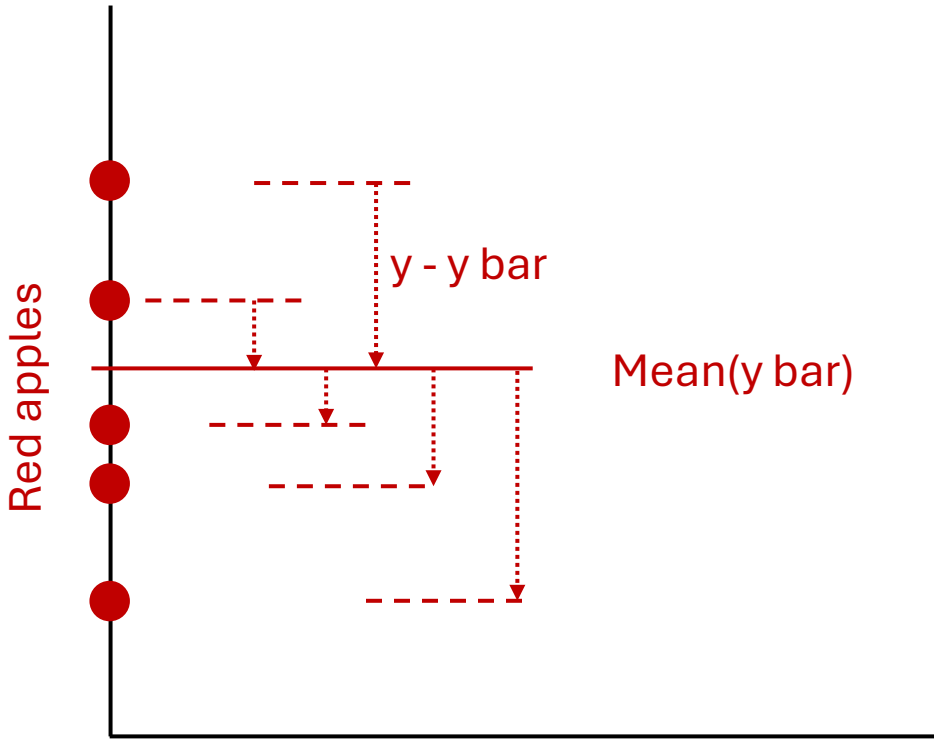
Covariance and correlation

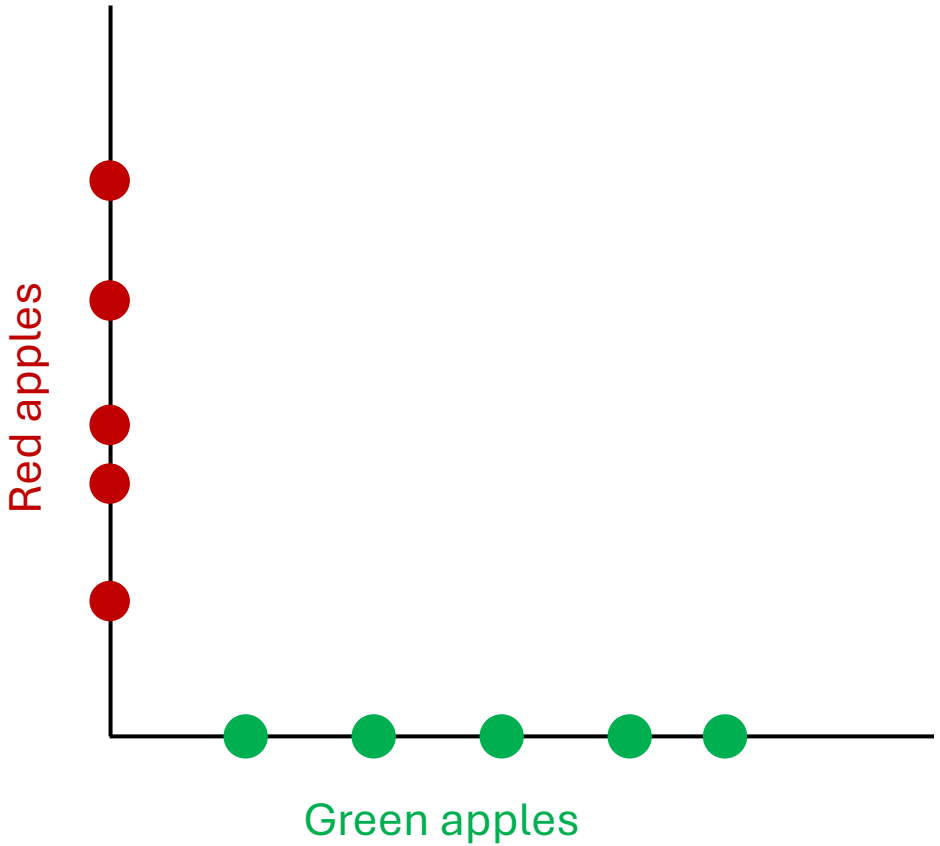
Kaung Myat Khant

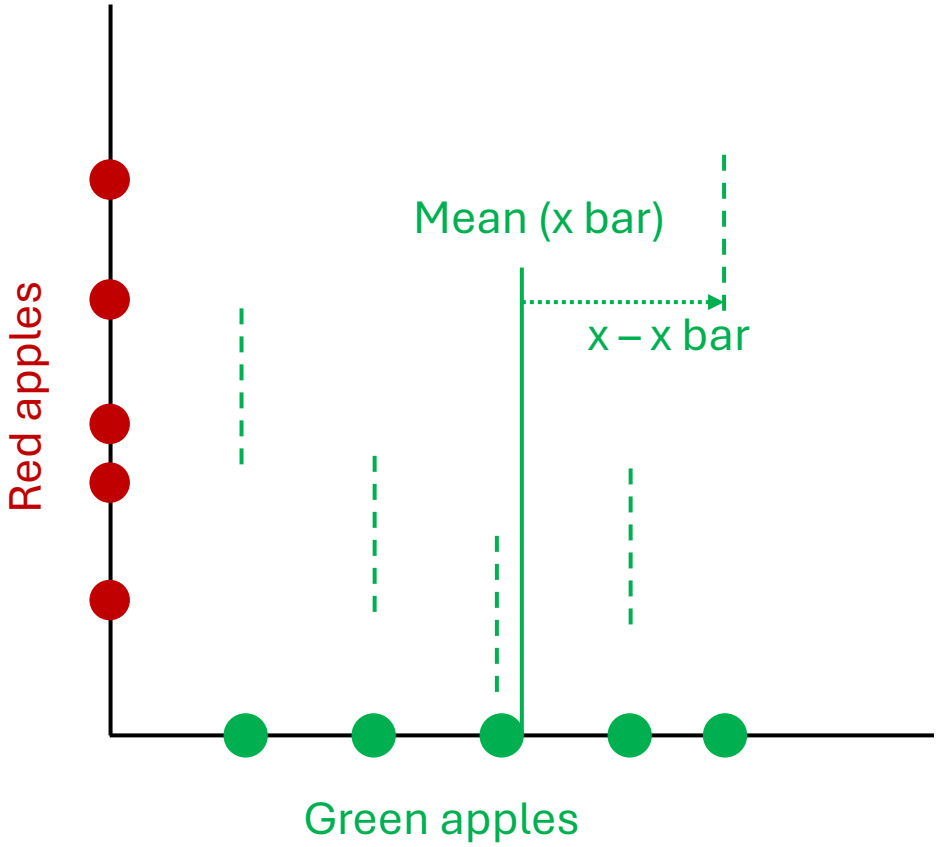



Red apples

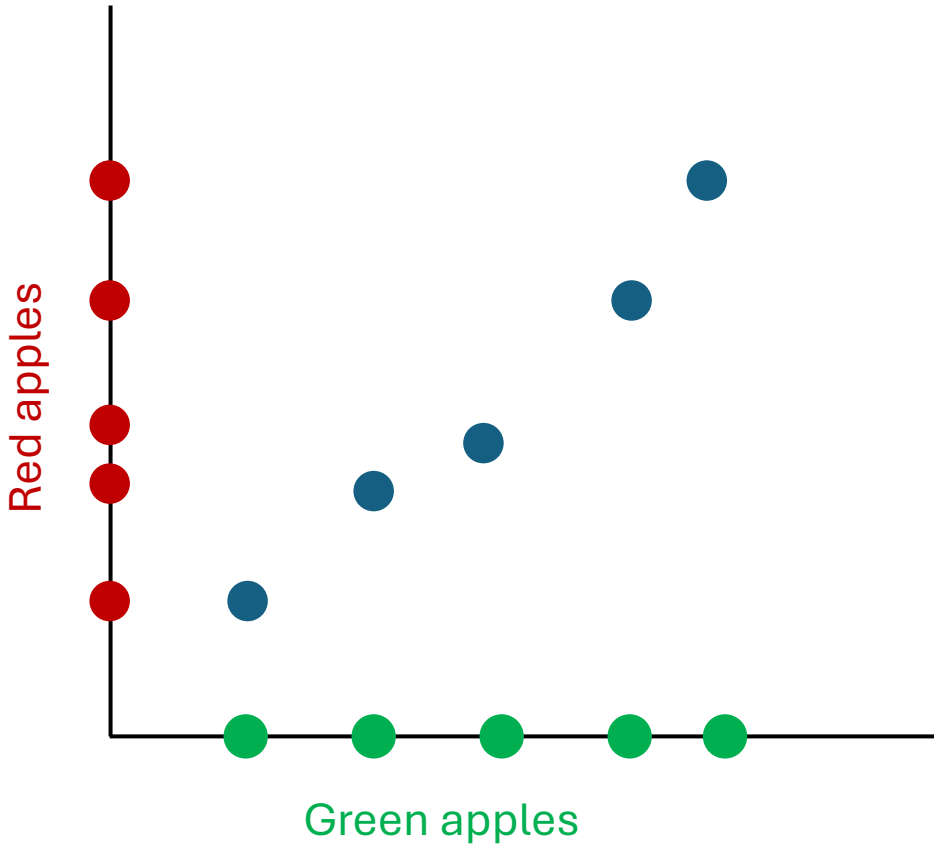


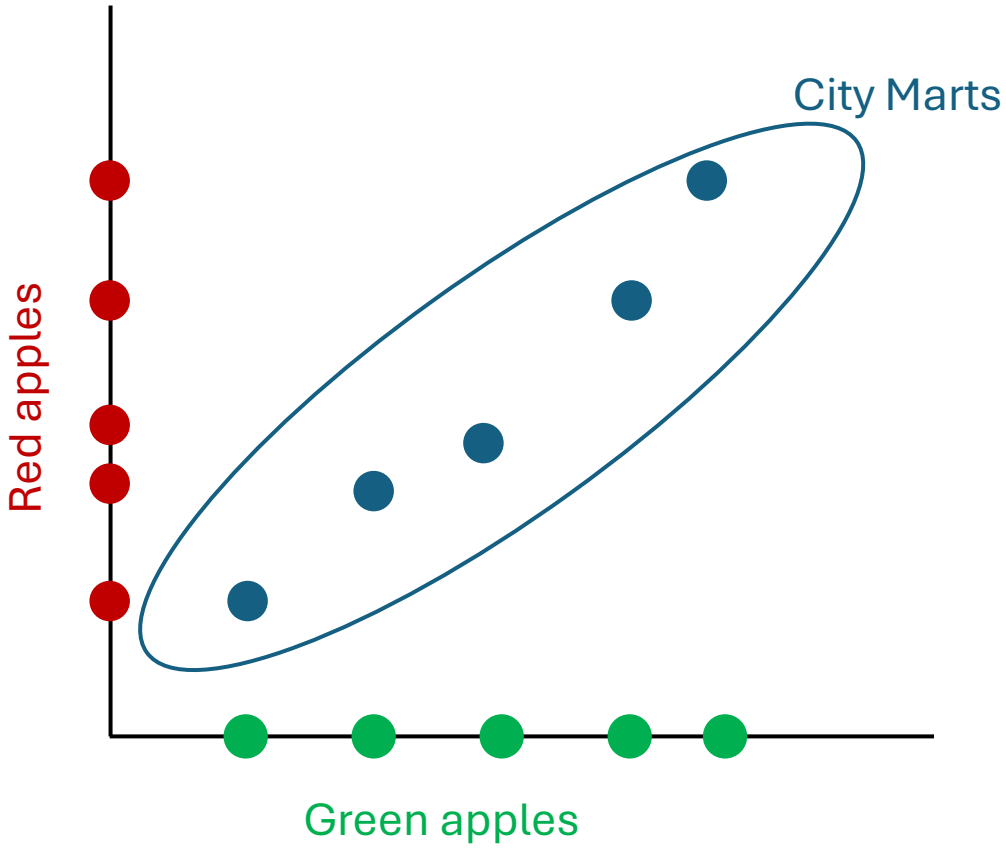
$$VarY = \frac{\sum(y - ybar)^2}{n - 1}$$

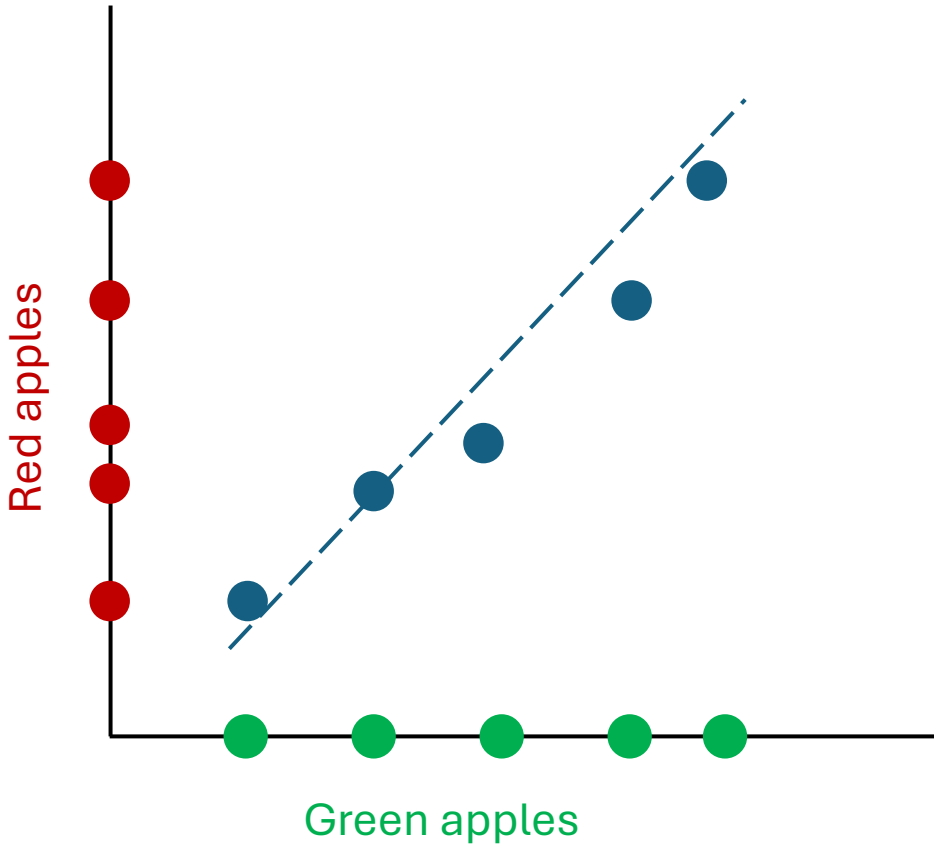

Red apples
Green apples



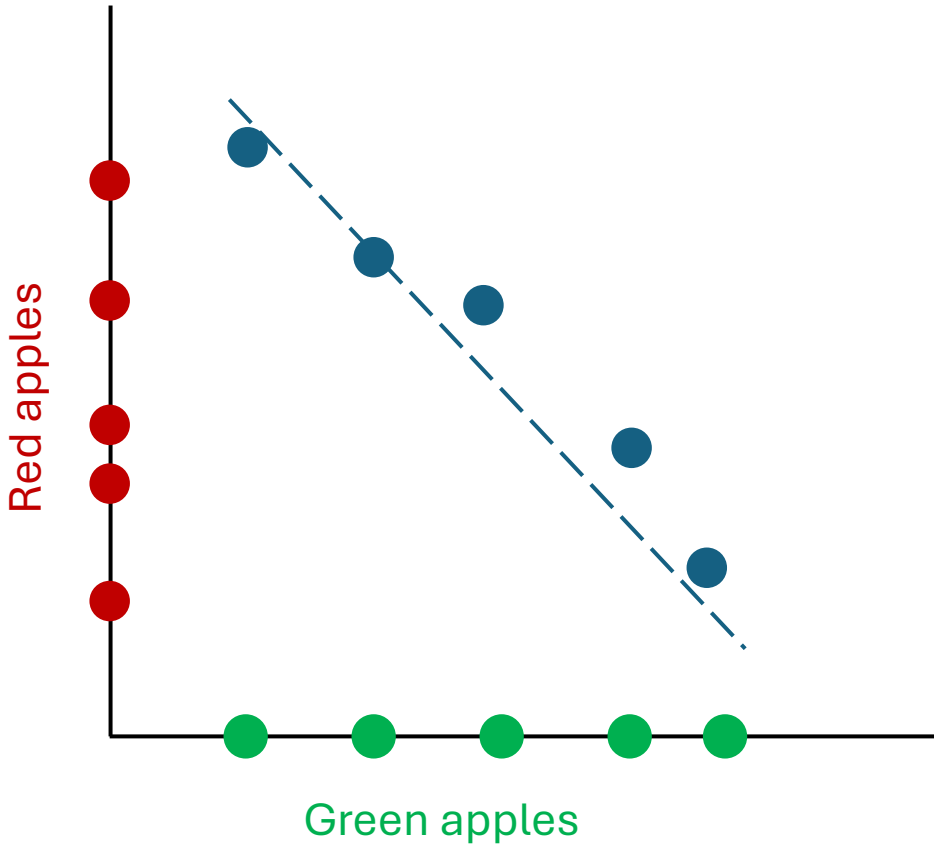
$$VarX = \frac{\sum(x\ - xbar)^2}{n-1}$$


Red apples
Green apples

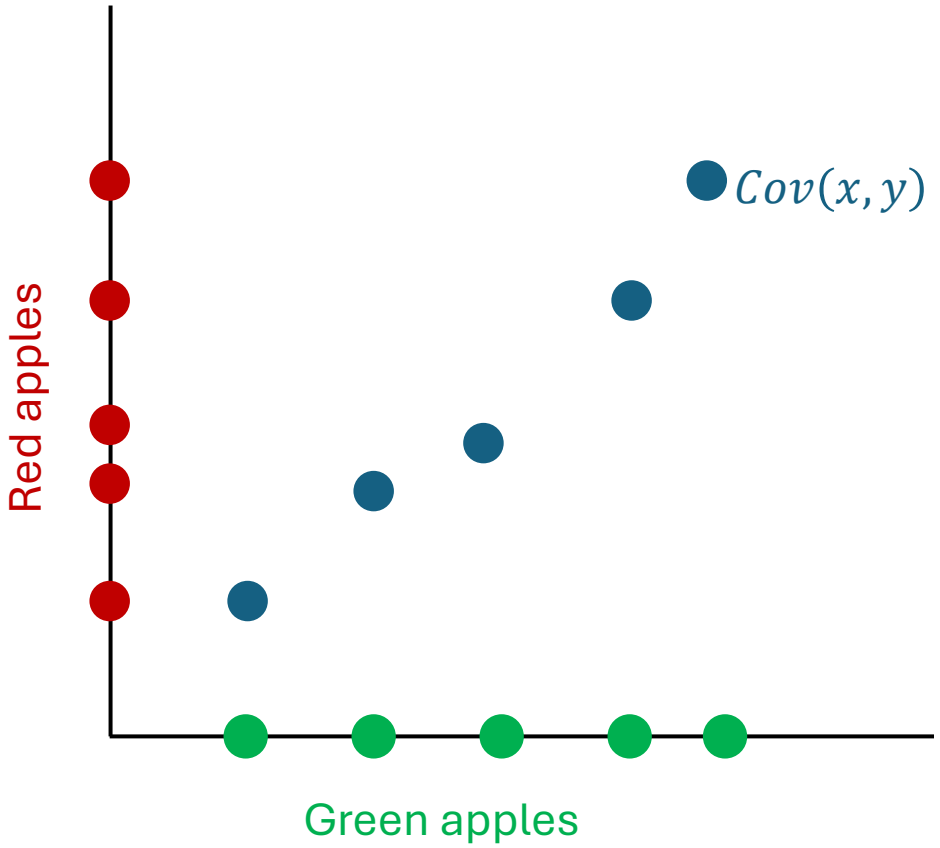

City Marts
Red apples
Green apples



Relationship - positive trend

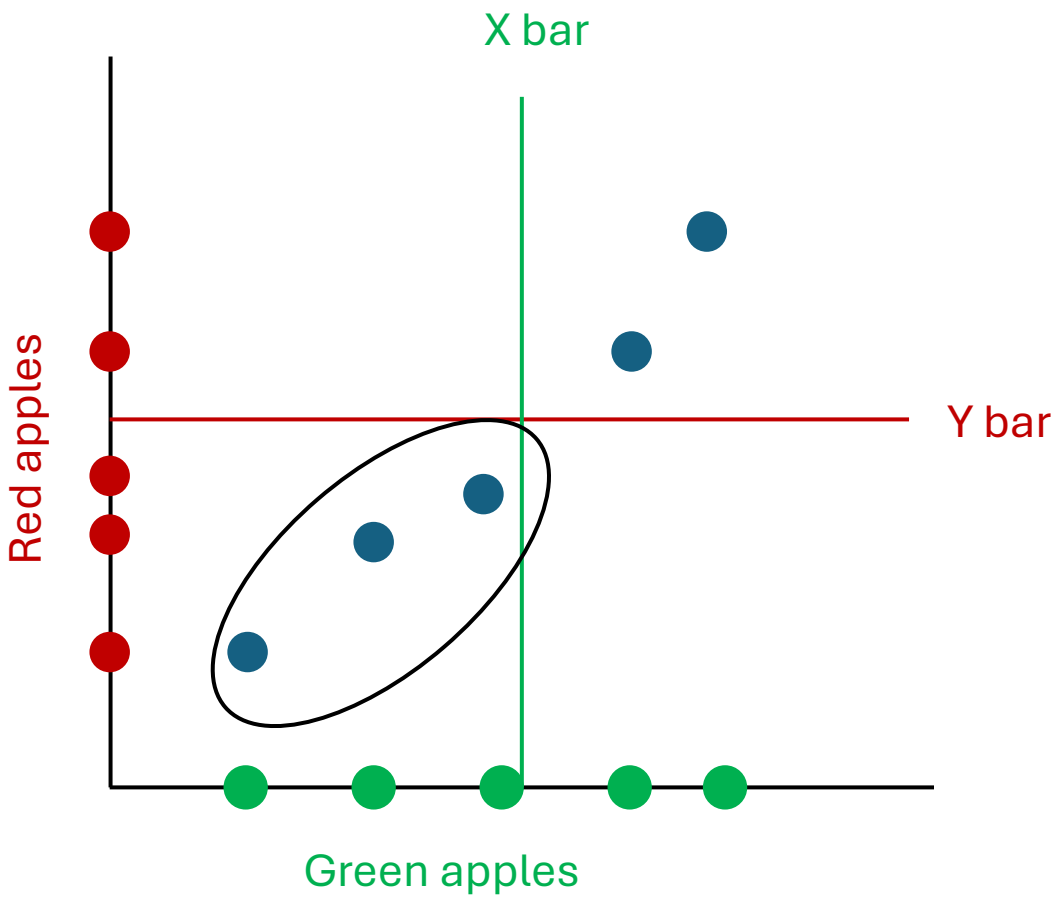


Relationship - negative trend



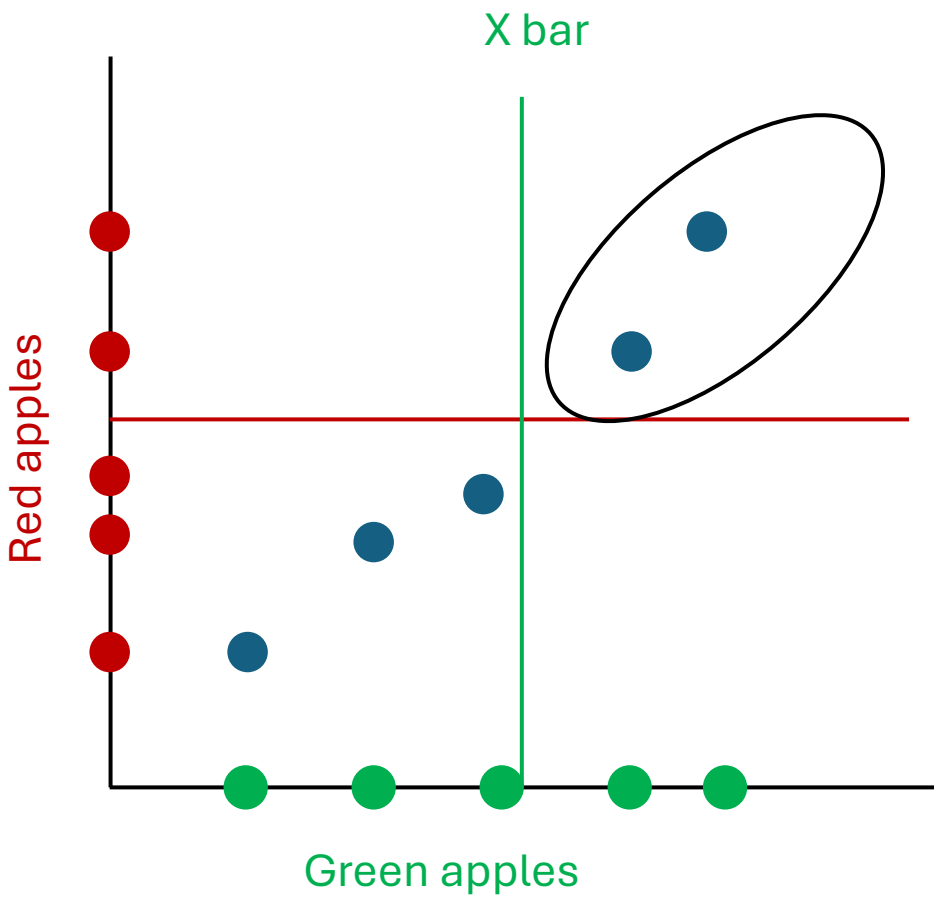
$$Cov(x,y) = \frac{\sum(x\ -xbar)(y\ -ybar)}{n-1}$$

$$Cov(x,y) = \frac{\sum(x\ - xbar)(y\ - ybar)}{n-1}$$



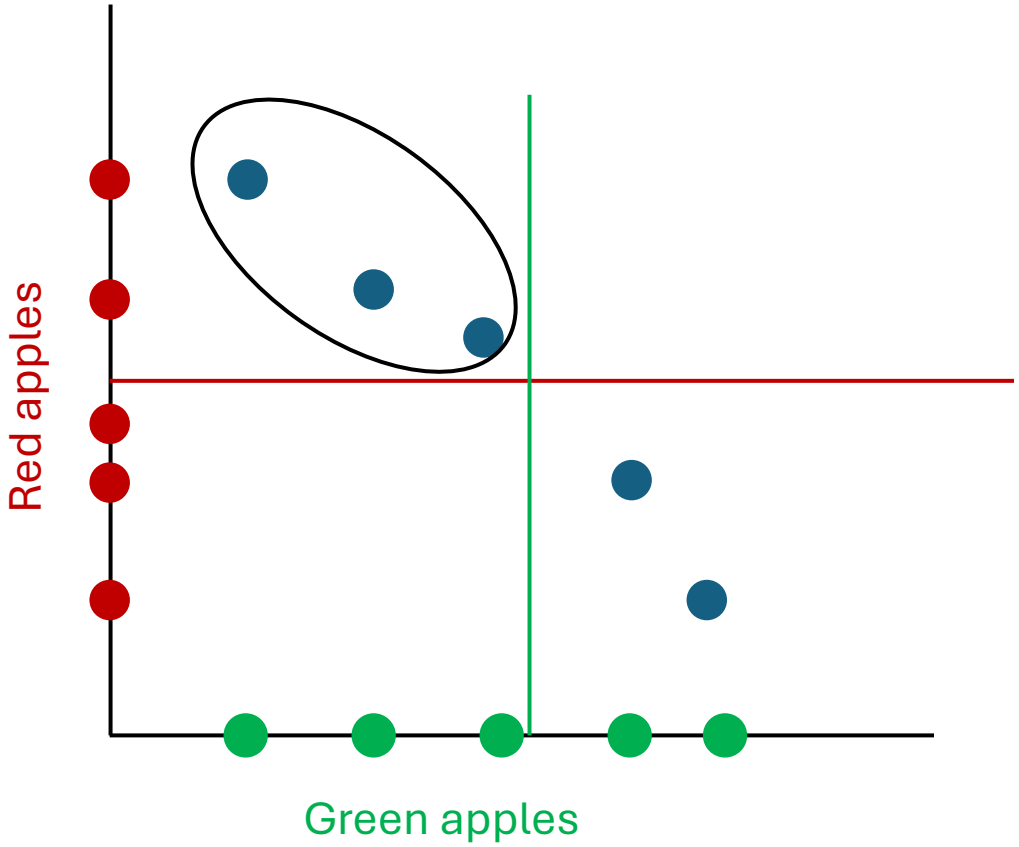
Negative x and negative y = positive

$$Cov(x,y) = \frac{\sum(x\ - xbar)(y\ - ybar)}{n-1}$$



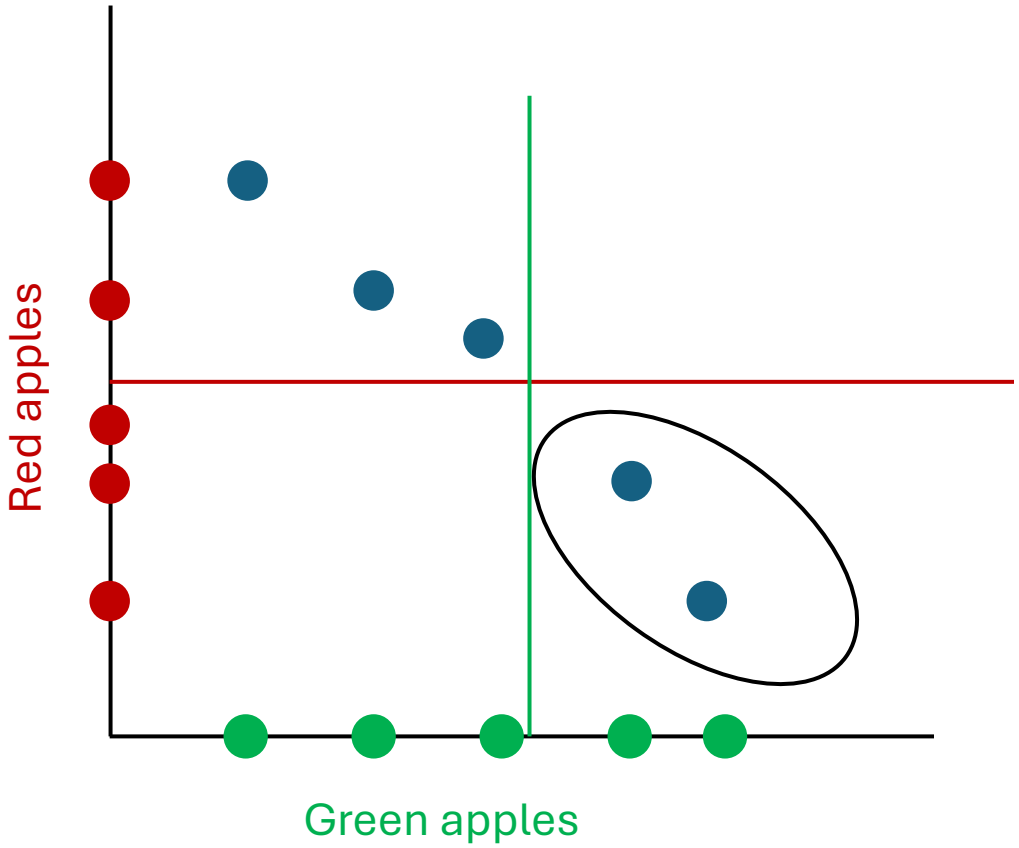
positive x and positive y = positive

$$Cov(x,y) = \frac{\sum(x\ - xbar)(y\ - ybar)}{n-1}$$



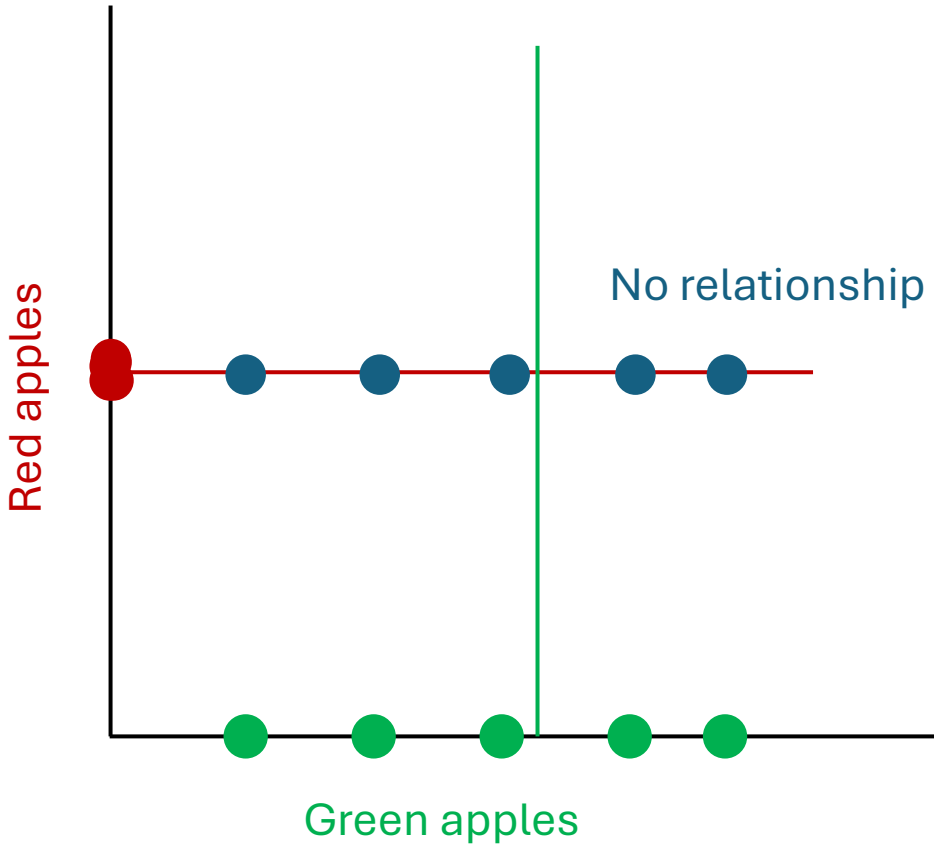
Negative x and positive y = negative

$$Cov(x,y) = \frac{\sum(x\ - xbar)(y\ - ybar)}{n-1}$$



positive x and negative y = negative

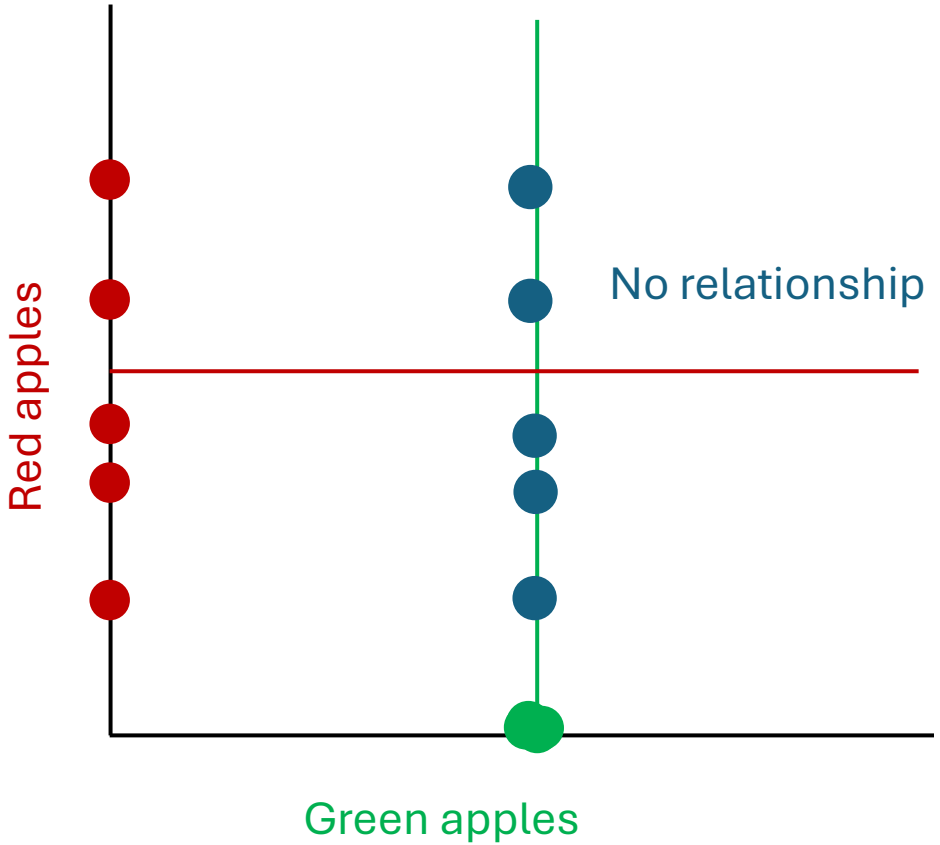
$$Cov(x,y) = \frac{\sum(x\ - xbar)(y\ - ybar)}{n-1}$$



negative x and zero y = zero

positive x and zero y = zero

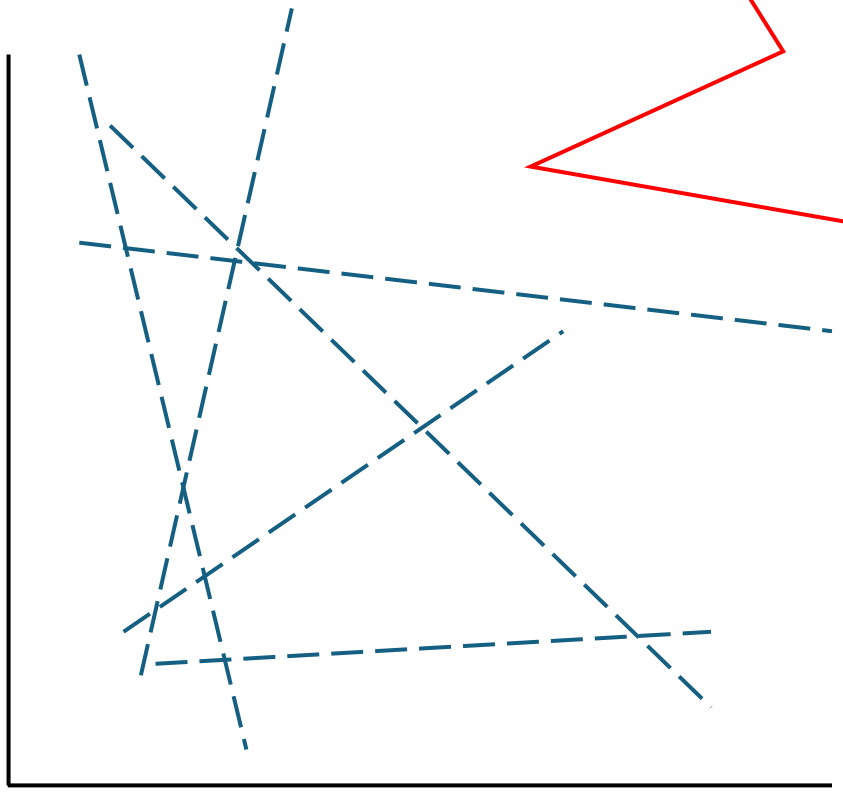
$$Cov(x,y) = \frac{\sum(x\ - xbar)(y\ - ybar)}{n-1}$$



Positive y and zero x = zero

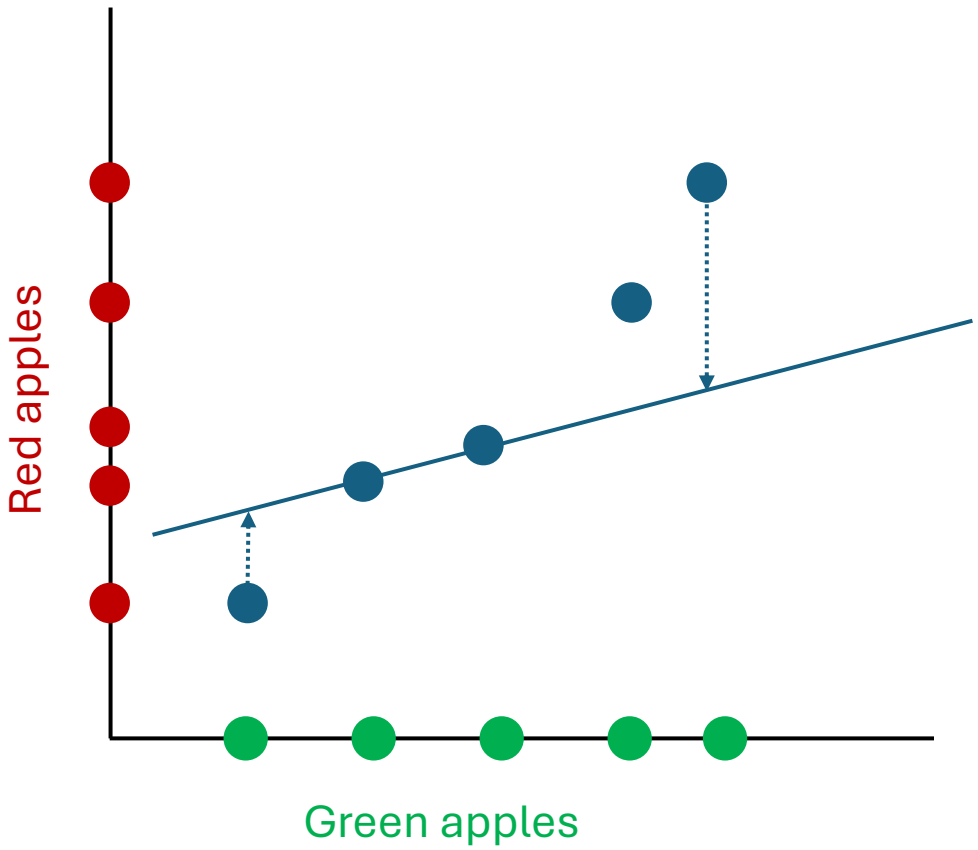
Negative y and zero x = zero

$$Cov(x, y) = \frac{\sum(x\ - xbar)(y\ - ybar)}{n - 1}$$



Cannot tell the slope

$$Cov(x,y) = \frac{\sum(x\ - xbar)(y\ - ybar)}{n-1}$$

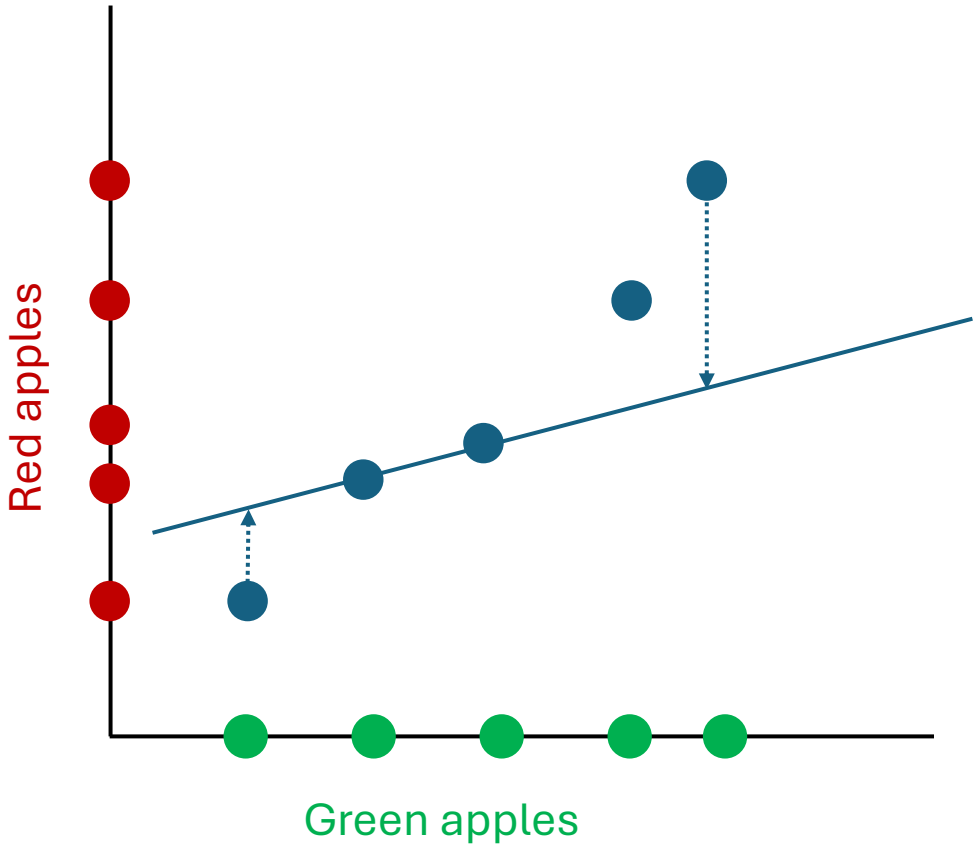


- Cannot tell the distance between the point and trend line
- Value depends on the scale
- Use to calculate correlation and blah blah blah

$$r = \frac{Cov(X,Y)}{\sqrt{VarX} \times \sqrt{VarY}}$$

Correlation

$$r = \frac{Cov(X,Y)}{\sqrt{VarX} \times \sqrt{VarY}}$$



- Two points can easily fit with a line
  - That is type 1 error
- Need larger sample size
- And correlation is not causation

Example 1

Observation	X	Y
1	1	2
2	2	5
3	3	7
4	4	8
5	5	8
Mean	3	6

i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})^2$	$(Y_i - \bar{Y})^2$	$(X_i - \bar{X})(Y_i - \bar{Y})$
1	−2	−4	4	16	+8
2	−1	−1	1	1	+1
3	0	+1	0	1	0
4	+1	+2	1	4	+2
5	+2	+2	4	4	+4
Sum (Σ)	0	0	10	26	15

VarX = 2.5; sdX = 1.5811;VarY = 6.5; sdY = 2.5495
Cov(X,Y) = 15/4 = 3.75
rho = 3.75/(1.58 x 2.54) = 0.93

Example 2

Observation	X	Y
1	10	2
2	20	5
3	30	7
4	40	8
5	50	8
Mean	30	6

i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})^2$	$(Y_i - \bar{Y})^2$	$(X_i - \bar{X})(Y_i - \bar{Y})$
1	−20	−4	400	16	+80
2	−10	−1	100	1	+10
3	0	+1	0	1	0
4	+10	+2	100	4	+20
5	+20	+2	400	4	+40
Sum (Σ)	0	0	1000	26	150

VarX = 250; sdX = 15.8114;VarY = 6.5; sdY = 2.5495

Cov(X,Y) = 150/4 = 37.5

rho = 37.5/(15.81 x 2.55) = 0.93

Example 3

Observation	X	Y
1	1	5
2	2	2
3	3	8
4	4	4
5	5	6
Mean	3	5

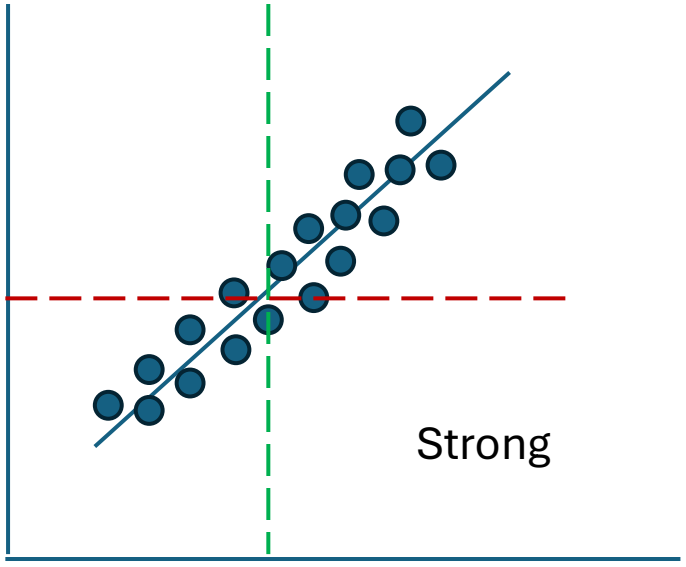
i	$Xi-\bar{X}$	$Yi-\bar{Y}$	$(Xi-\bar{X})2$	$(Yi-\bar{Y})2$	$(Xi-\bar{X})(Yi-\bar{Y})$
1	−2	0	4	0	0
2	−1	−3	1	9	+3
3	0	+3	0	9	0
4	+1	−1	1	1	-1
5	+2	+1	4	1	+2
Sum (Σ)	0	0	10	20	4

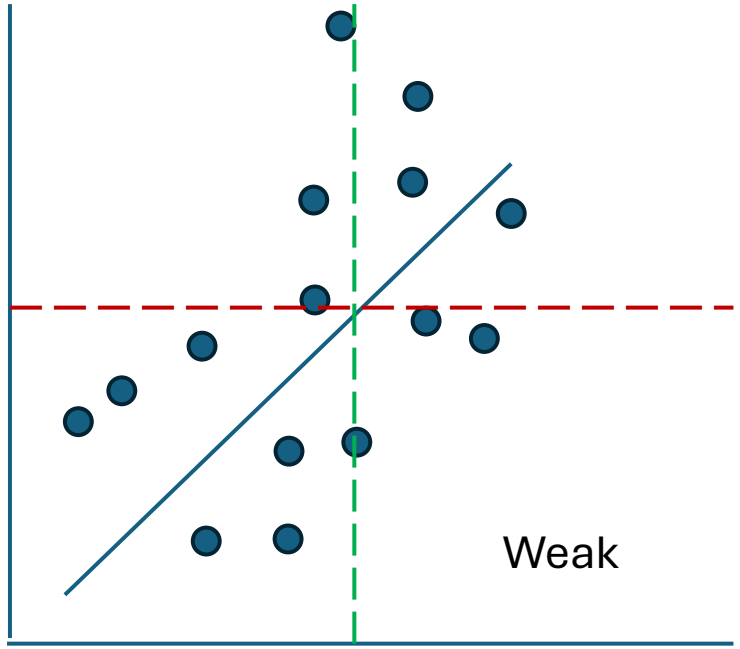
VarX = 2.5; sdX = 1.5811;VarY = 5; sdY = 2.2361

Cov(X,Y) = 4/4 = 1

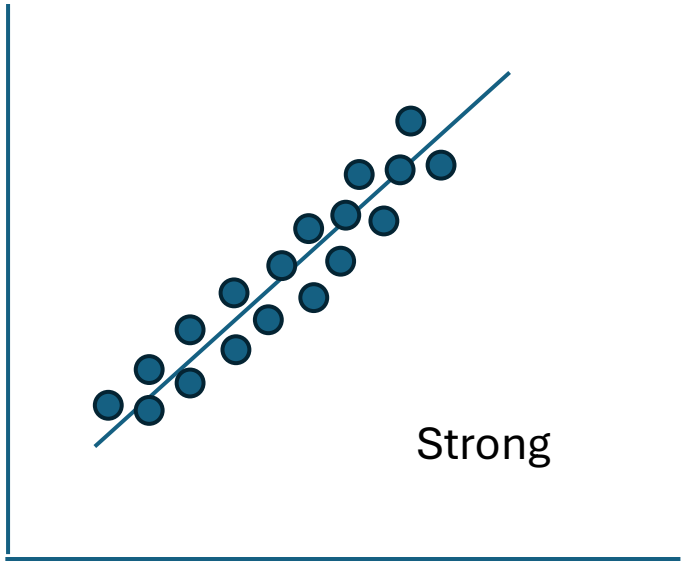
rho = 1/(1.58 x 2.24) = 0.28

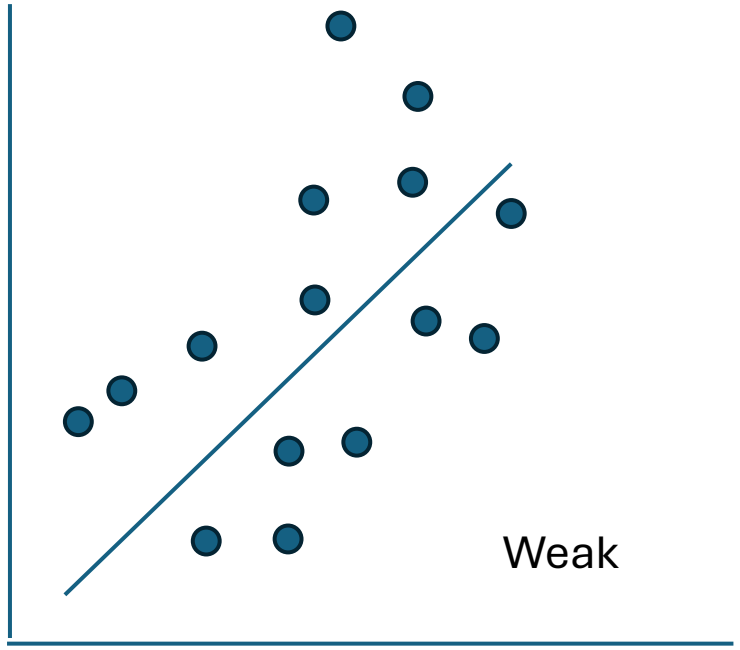
$$r = \frac{Cov(X,Y)}{\sqrt{VarX} \times \sqrt{VarY}}$$



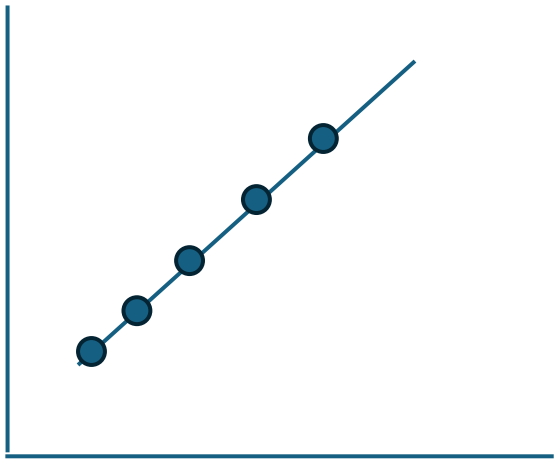


$$r = \frac{Cov(X,Y)}{\sqrt{VarX} \times \sqrt{VarY}}$$

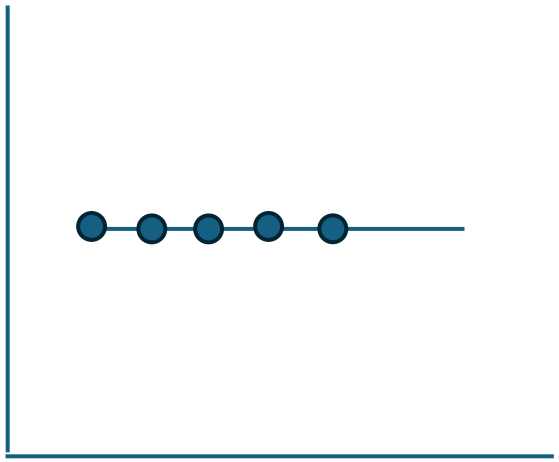




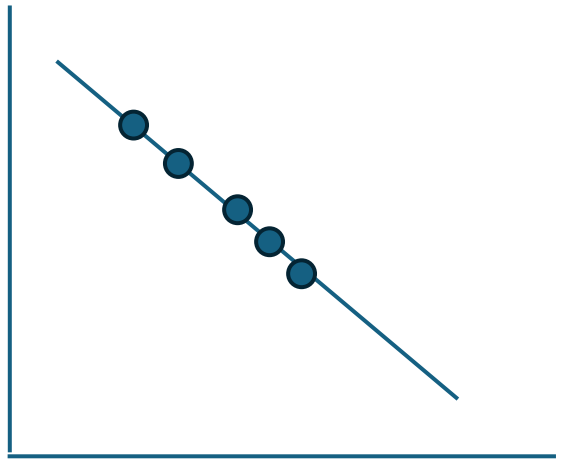
$$r = \frac{Cov(X,Y)}{\sqrt{VarX} \times \sqrt{VarY}}$$



r = 1



r = 0



r = -1

- Two points can easily fit with a line (meaningless)
  - That is type 1 error
- Need larger sample size for more confident
- And correlation is not causation
- Still cannot tell the slope

Hypothesis testing

- Null : r = 0
- $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$
- df = n - 2